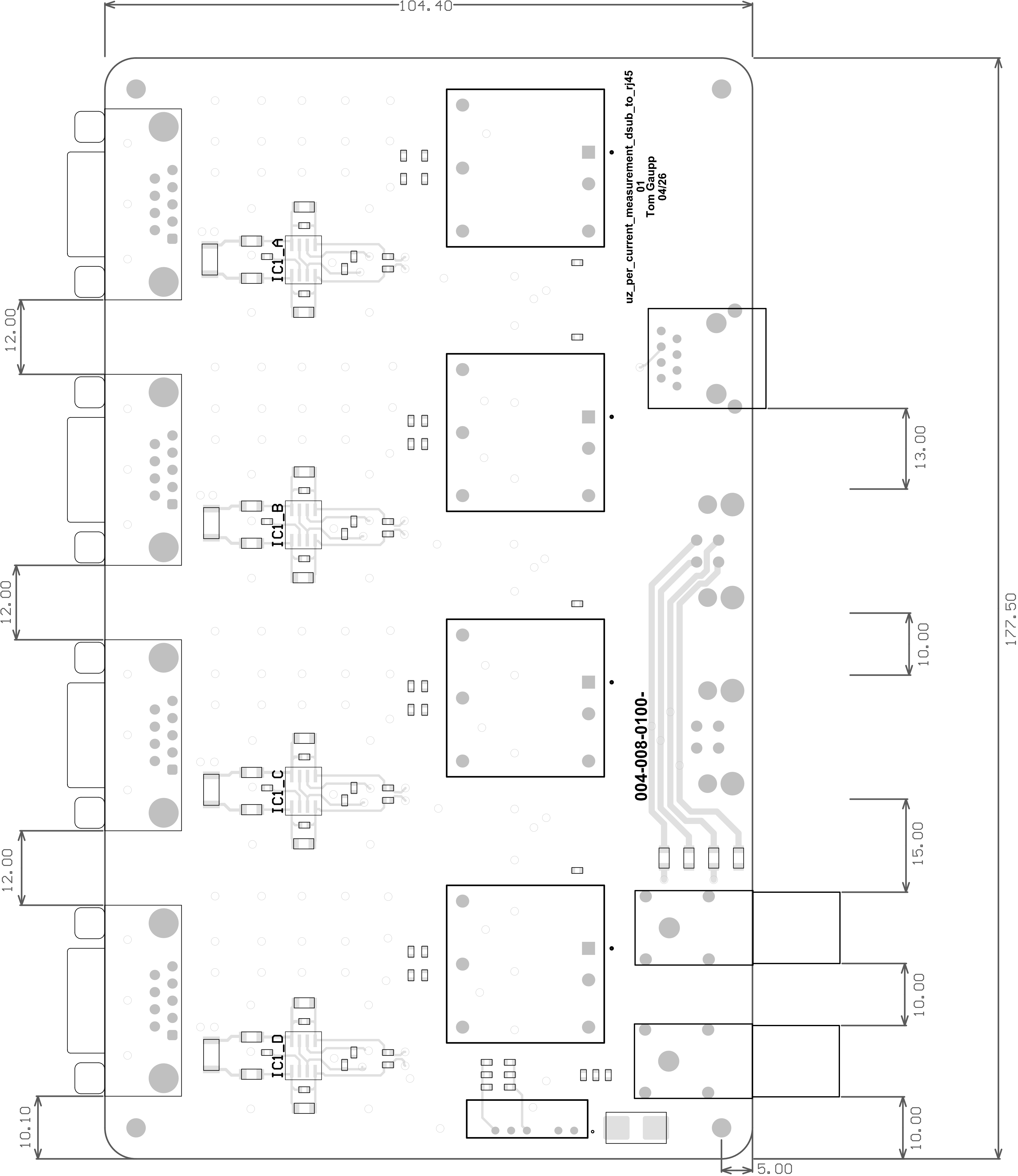
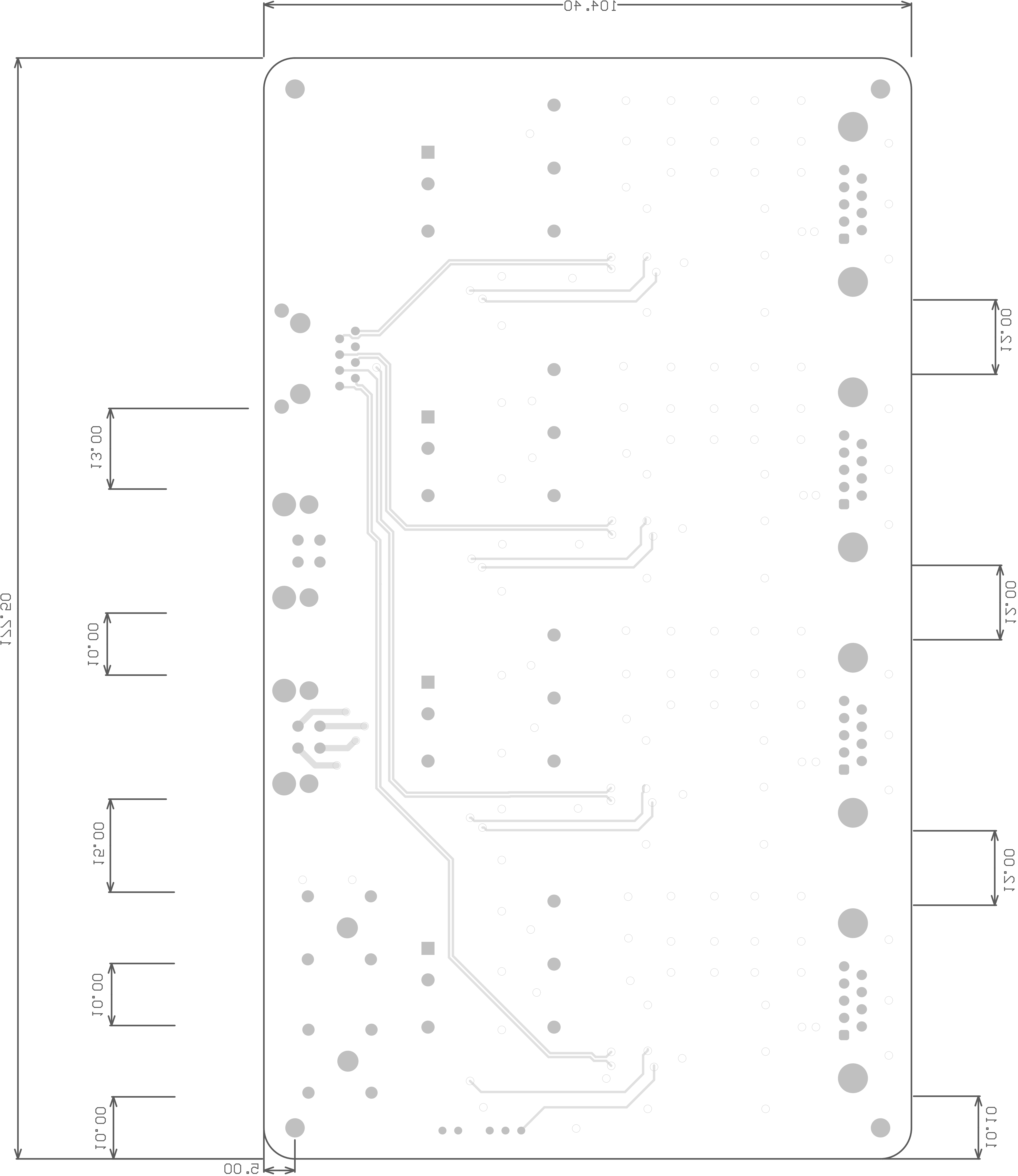


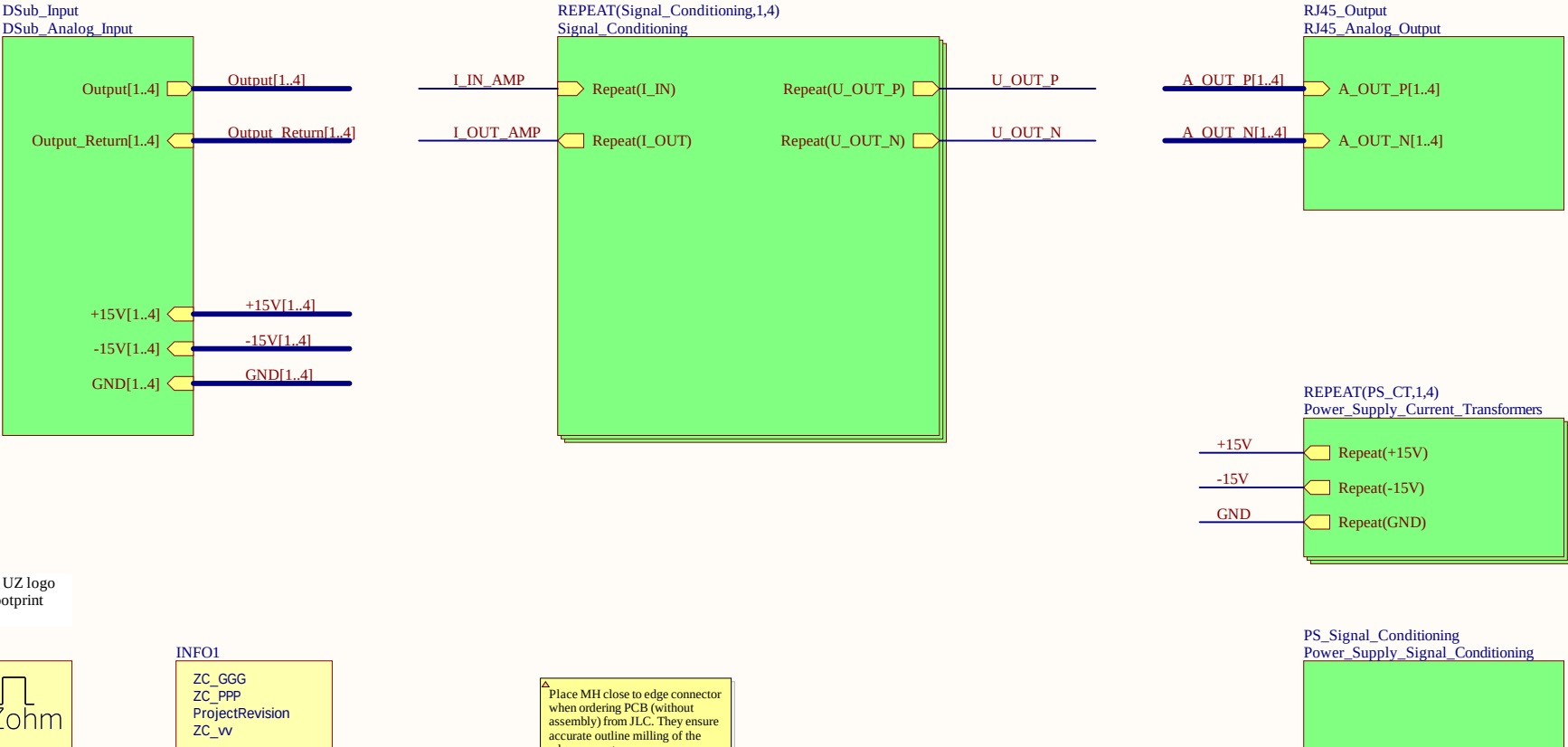




TopSheet

|                |            |
|----------------|------------|
| Output1        | I_IN_AMP1  |
| Output_Return1 | I_OUT_AMP1 |
| Output2        | I_IN_AMP2  |
| Output_Return2 | I_OUT_AMP2 |
| Output3        | I_IN_AMP3  |
| Output_Return3 | I_OUT_AMP3 |
| Output4        | I_IN_AMP4  |
| Output_Return4 | I_OUT_AMP4 |

|          |          |
|----------|----------|
| U_OUT_P1 | A_OUT_P1 |
| U_OUT_N1 | A_OUT_N1 |
| U_OUT_P2 | A_OUT_P2 |
| U_OUT_N2 | A_OUT_N2 |
| U_OUT_P3 | A_OUT_P3 |
| U_OUT_N3 | A_OUT_N3 |
| U_OUT_P4 | A_OUT_P4 |
| U_OUT_N4 | A_OUT_N4 |



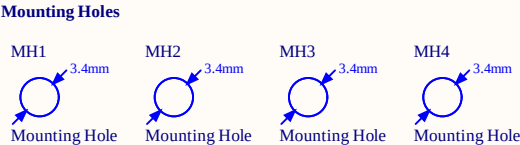
Chose size of UZ logo  
by chosing footprint  
type



INFO1  
ZC\_GGG  
ZC\_PPP  
ProjectRevision  
ZC\_vv  
ZC Serialnumber

Place MH close to edge connector  
when ordering PCB (without  
assembly) from JLC. They ensure  
accurate outline milling of the  
edge connector.

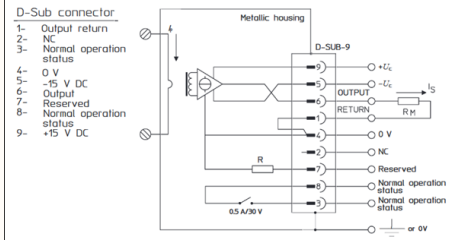
INFO2  
Project  
ProjectRevision  
AuthorParam  
ProjectDate  
Design Information



|                                                         |                            |                                                                              |  |
|---------------------------------------------------------|----------------------------|------------------------------------------------------------------------------|--|
| Title TopSheet.SchDoc                                   |                            | <b>UltraZohm</b><br><a href="http://www.ultrazohm.com">www.ultrazohm.com</a> |  |
| Revision: 01                                            | Design Engineer: Tom Gaupp |                                                                              |  |
| Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb |                            |                                                                              |  |
| Date: 03/06/2026<br>Sheet 1 of 12                       |                            |                                                                              |  |



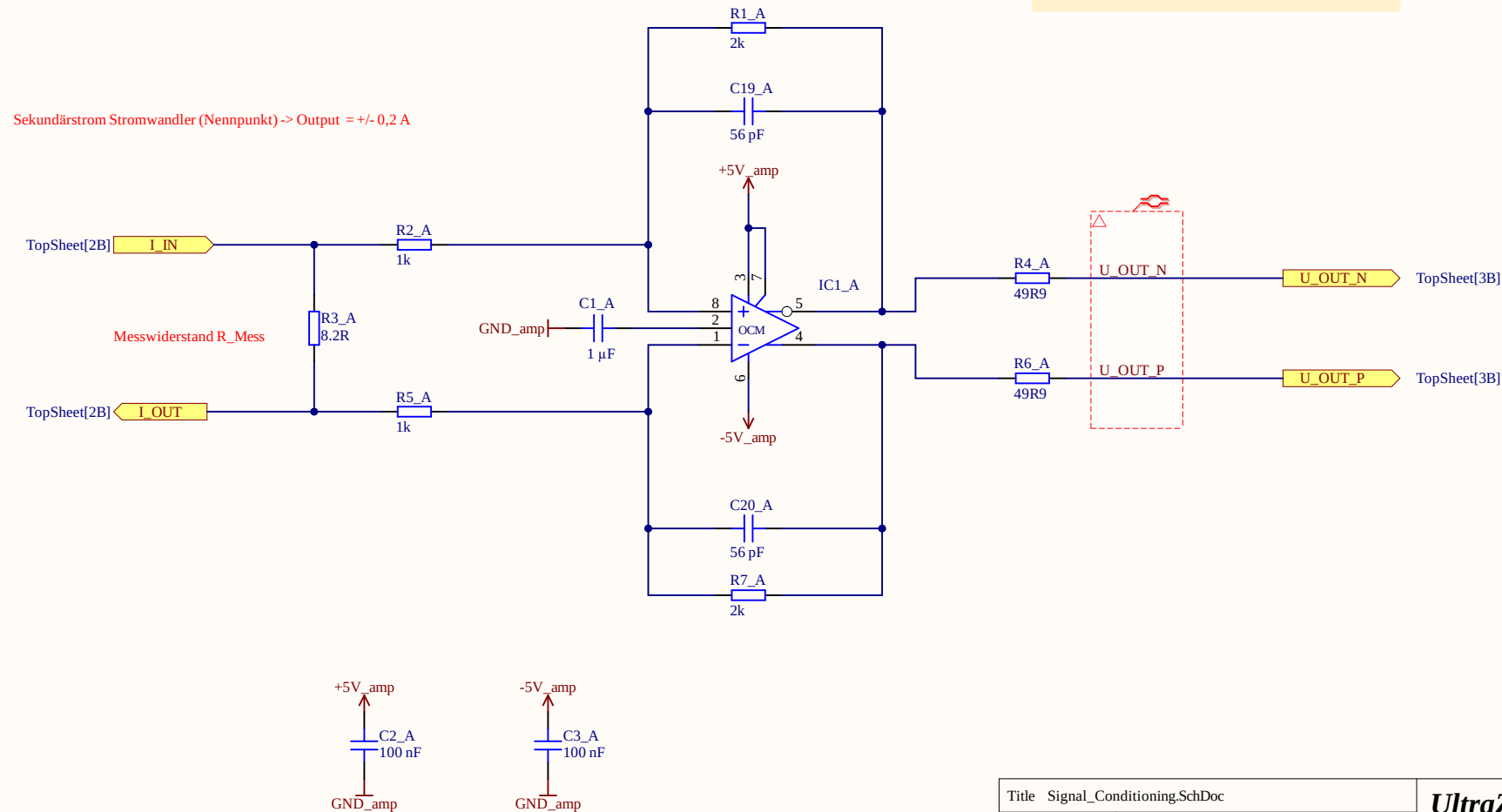
## Connection



## Signal Conditioning 1.4

Verstärkerschaltung Gain:  $G_{amp} = 2$  $G_{amp} = R_F / R_G$ 

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz:  $f_g$ Rückkopplungskondensator:  $C_f =$  $C_f = 1 / (2 \cdot \pi \cdot R_f \cdot f_g) = \dots =$ 

Title Signal\_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

UltraZohm

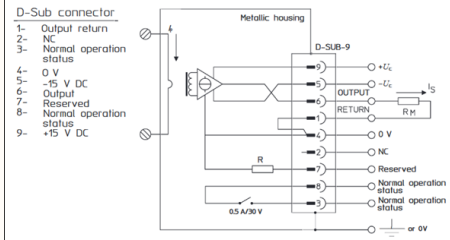
[www.ultrazohm.com](http://www.ultrazohm.com)

Date: 03/06/2026

Sheet 2.1 of 12



## Connection



# Signal Conditioning 1.4

Verstärkerschaltung Gain:  $G_{amp} = 2$

$G_{amp} = R_F / R_G$

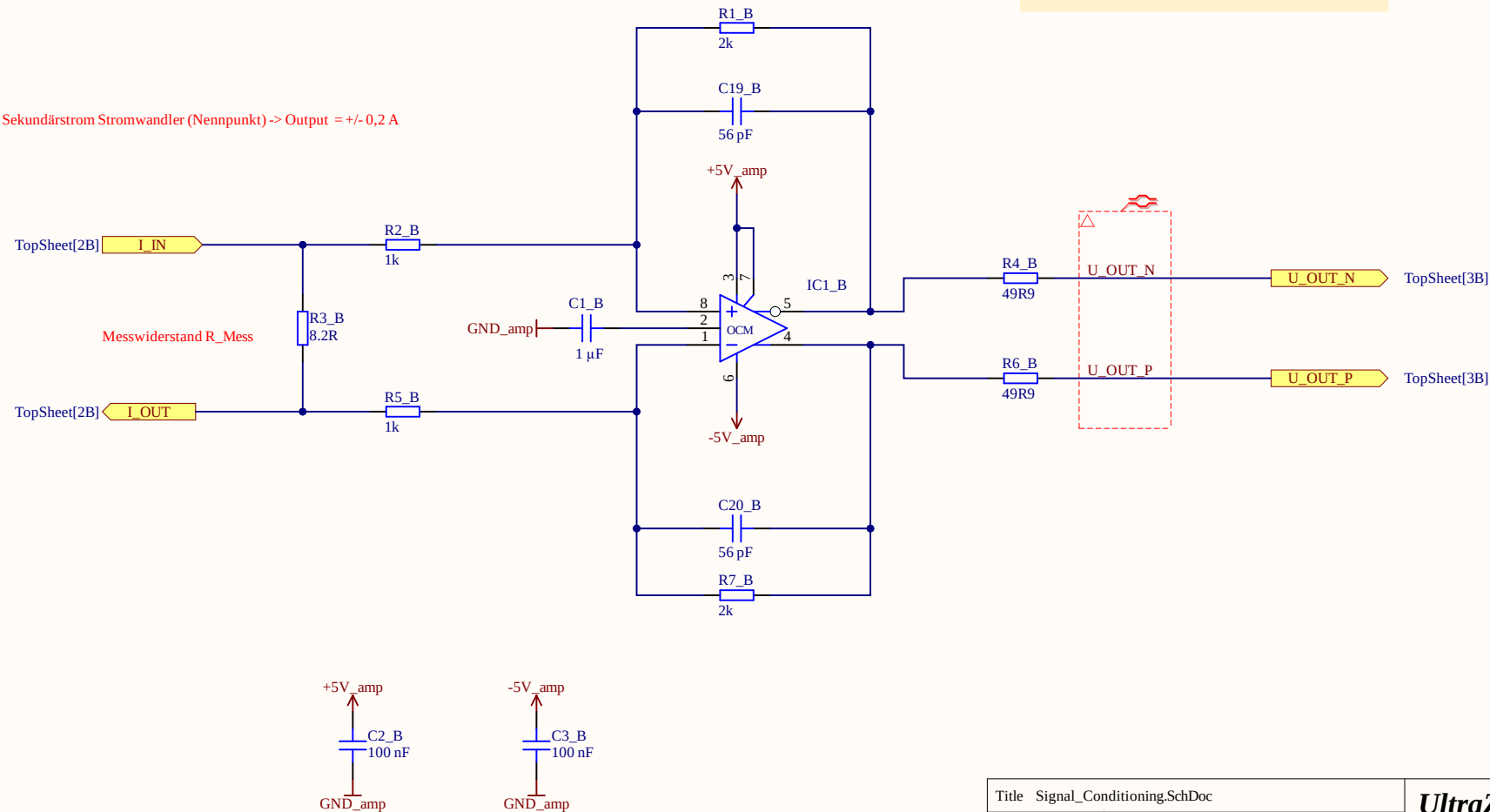
Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz:  $f_g$

Rückkopplungskondensator:  $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$

Sekundärstrom Stromwandler (Nennpunkt)  $\rightarrow$  Output =  $\pm 0,2$  A



Title Signal\_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

**UltraZohm**

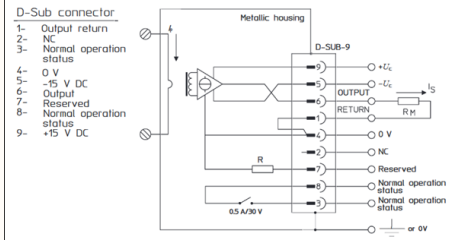
[www.ultrazohm.com](http://www.ultrazohm.com)

Date: 03/06/2026

Sheet 2.2 of 12



## Connection



# Signal Conditioning 1.4

Verstärkerschaltung Gain:  $G_{amp} = 2$

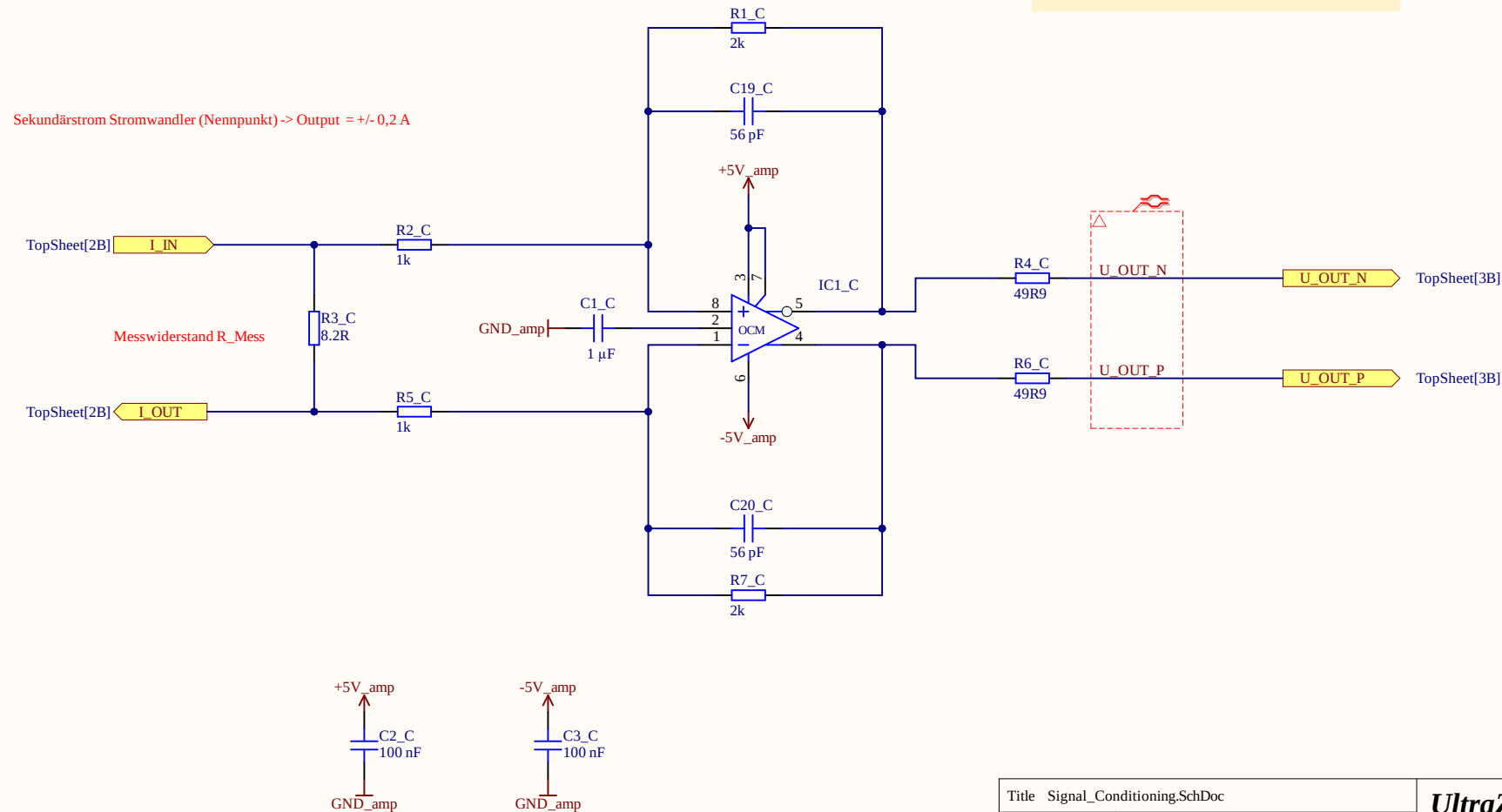
$G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz:  $f_g$

Rückkopplungskondensator:  $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$



Title Signal\_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

**UltraZohm**

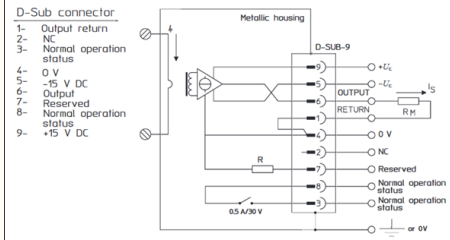
[www.ultrazohm.com](http://www.ultrazohm.com)

Date: 03/06/2026

Sheet 2.3 of 12



## Connection



# Signal Conditioning 1.4

Verstärkerschaltung Gain:  $G_{amp} = 2$

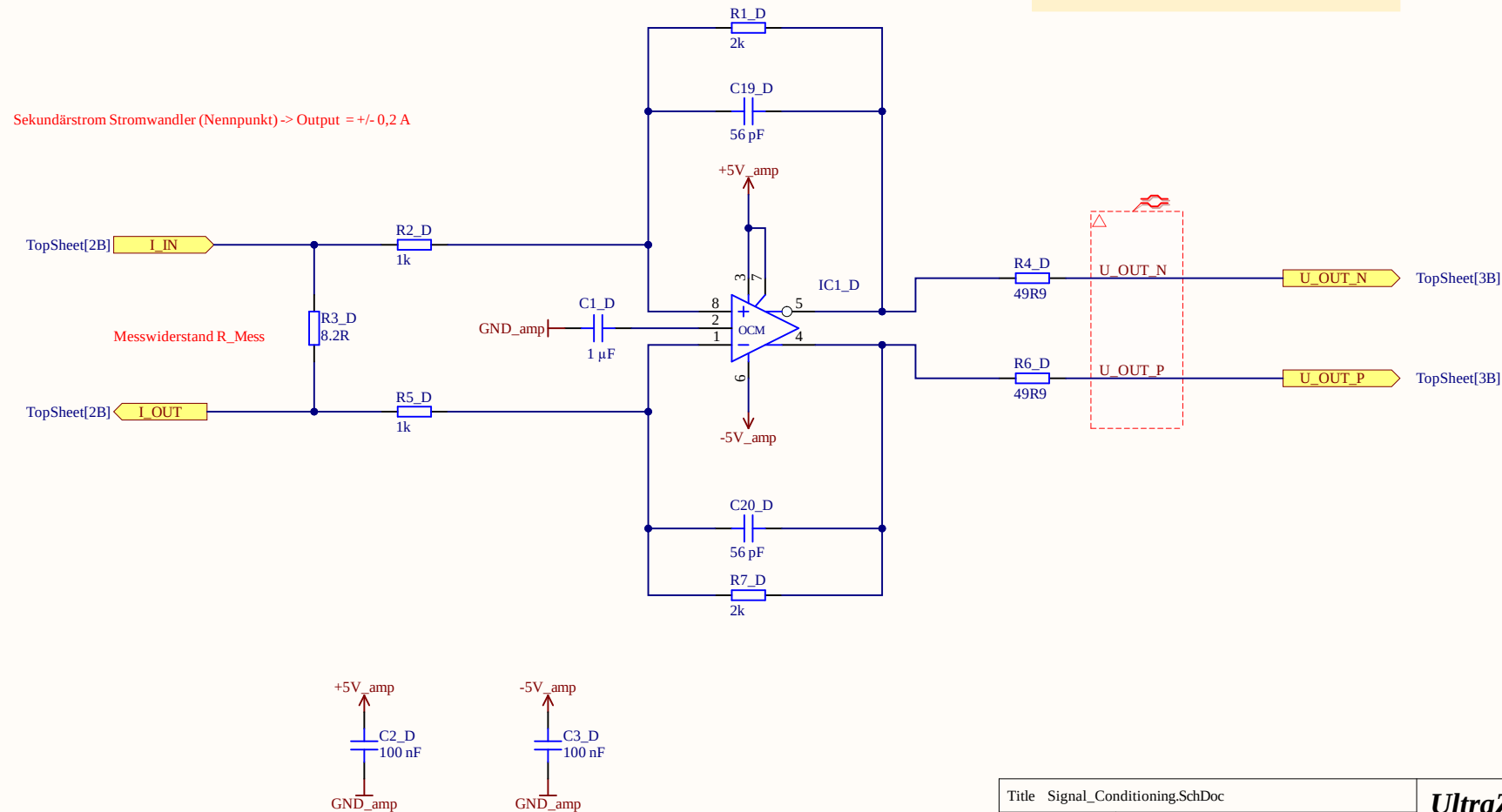
$G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz:  $f_g$

Rückkopplungskondensator:  $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$



Title Signal\_Conditioning.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

**UltraZohm**

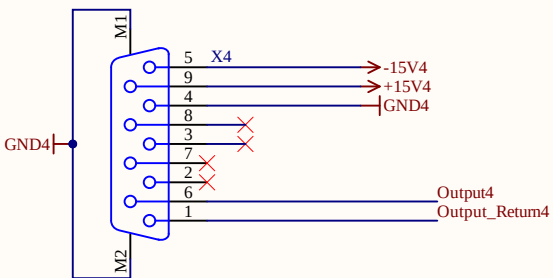
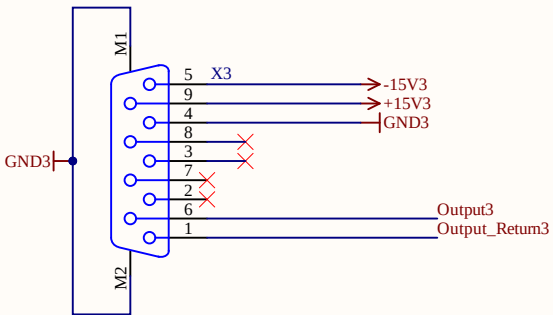
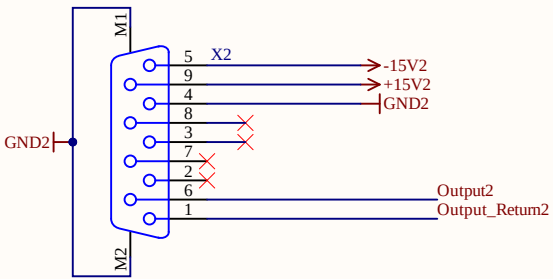
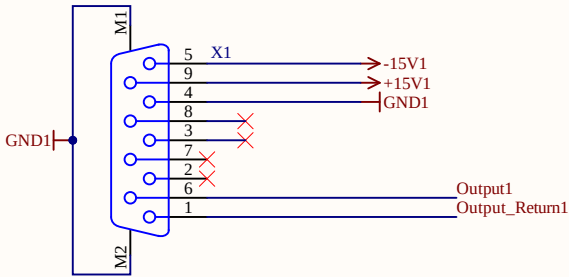
[www.ultrazohm.com](http://www.ultrazohm.com)

Date: 03/06/2026

Sheet 2.4 of 12



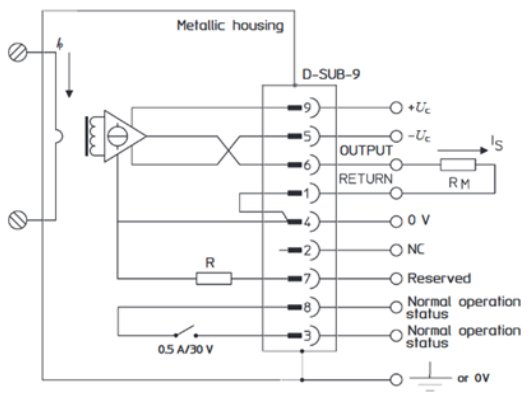
# Analog Input



## Connection

### D-Sub connector

- 1- Output return
- 2- NC
- 3- Normal operation status
- 4- 0 V
- 5- -15 V DC
- 6- Output
- 7- Reserved
- 8- Normal operation status
- 9- +15 V DC



Output[1..4]      Output[1..4]      TopSheet[1B]

Output\_Return[1..4]      Output\_Return[1..4]      TopSheet[1B]

+15V[1..4]      +15V[1..4]      TopSheet[1B]

-15V[1..4]      -15V[1..4]      TopSheet[1C]

GND[1..4]      GND[1..4]      TopSheet[1C]

Title DSub\_Analog\_Input.SchDoc

Revision: 01      Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

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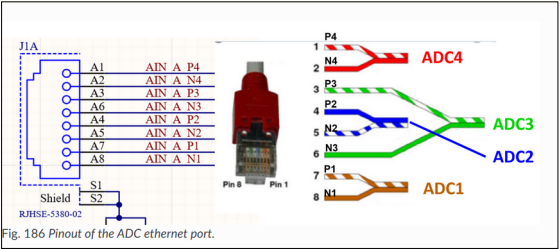
Date: 03/06/2026

Sheet 3 of 12





# Analog Output





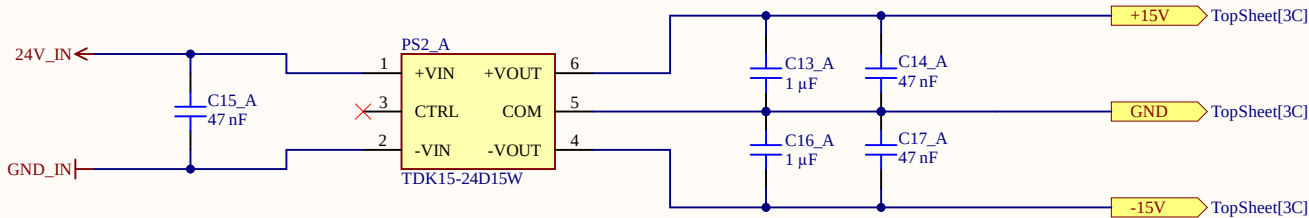
# Power Supply Currentsensors

LEM IN 100-S

Spannungsversorgung: +/- 15 V  
DC Nennstromaufnahmen @ 15V = 320 mA  
I\_max = 420 mA

Secondary nominal RMS current (@100A) = 0,2 A  
Shielded output cable & shielded plug are recommended to use

Max Power Consumption (pro Stromsensor):  
P = U\*I\_max = 15 V \* 0,420 A = 6,3 W



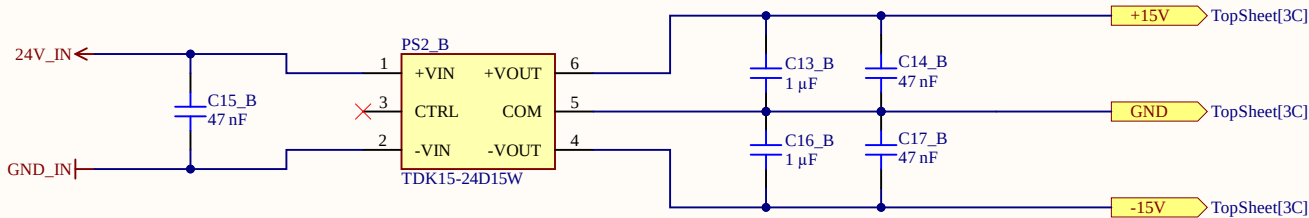
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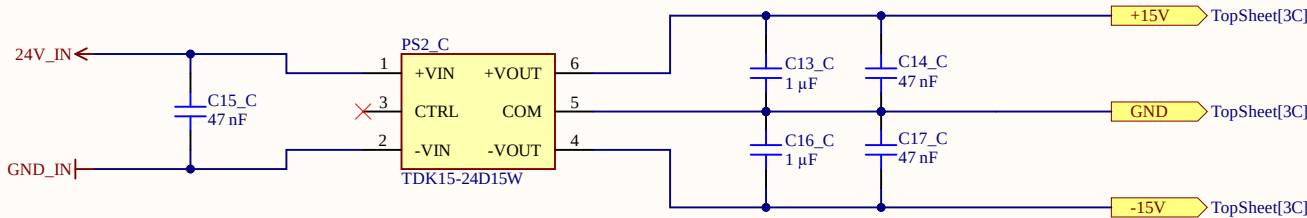
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